PHYS1201 Cheat Sheet

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1 Classical Physics

Lagrange

$$L = K - V$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}} = \frac{\partial L}{\partial x}$$

Cyclic variables are variables which don't appear in the Lagrangian.

Hamiltonian and Stationary action

$$H = \sum_i p_i \dot{q}_i - L$$

Where q is all cyclic variables.

$$S = \int_{t_1}^{t_2} L dt$$

$$\delta S = 0; \frac{dS}{dt} = 0$$

$S \gg \hbar$ means that the paths get cut down a lot, to the point where only the stationary action survives.

2 Quanta

Equations

$$\varphi_{A\to B} = N \sum_{\text{paths}} e^{iS/\hbar}$$

$$P = |\varphi_{A\to B}|^2$$

Misc

$$\hbar = \frac{h}{2\pi}$$

$$\lambda = \frac{h}{p}$$

Stationary Action

$$S = 2\pi n\hbar = nh$$

3 Special Relativity

Dilation

$$\gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}$$

$$x' = \gamma(x - vt)$$

$$t' = \gamma\left(t - \left(\frac{xv}{c^2}\right)\right)$$

$$\Delta t' = \gamma \Delta t$$

$$\Delta x' = \frac{\Delta x}{\gamma}$$

Relativistic Equations

$$E = \gamma mc^2 = \sqrt{m^2c^4 + p^2c^2}$$

$$p = \gamma mv$$

4 Particles

$$Q = (\Delta M_{\text{Reactants}} - \Delta M_{\text{Products}})c^2$$

Forbidden if Q value is less than 0

$$T_{1/2} = \tau \ln(2)$$

Same form for oscillators: amplitude has $\tau_A = \frac{2}{\gamma}$, energy has $\tau_E = \frac{1}{\gamma}$, so $T_{1/2} = \tau \ln(2)$ with the matching τ.

5 Oscillation

General Form

$$m\ddot{x} + r\dot{x} + kx = 0$$

$$\ddot{x} + \gamma\dot{x} + \omega_0^2 x = 0$$

$$\omega_0^2 = \frac{g}{L}$$

$$\omega = 2\pi f$$

- The period of an undamped pendulum: $T = 2\pi\sqrt{\frac{L}{g}}$

$$E = E_0 e^{-\gamma t}$$

$$A = A_0 e^{-\frac{\gamma t}{2}}$$

$$\tau = \frac{2}{\gamma}$$

Max amplitude at $\omega = \sqrt{\omega_0^2 - \frac{\gamma^2}{2}}$ But max velocity and power at ω_0

Driven Oscillator

$$m\ddot{x} + r\dot{x} + kx = F_0 e^{i\omega t}$$

$$Ae^{i\varphi} = \frac{F_0}{m(\omega_0^2 - \omega^2 + i\gamma\omega)}$$

$$Q = \frac{\omega_0}{\Delta w} = \frac{\omega_0}{\gamma}$$

- Δw is the frequency difference between the two halfway points of the peak

Couple Oscillator

- The symmetric frequency is when the the vector has equal components, so $\binom{1}{1}$ and the antisymmetric node would be $\binom{1}{-1}$

Impedance

$$Z = \frac{m}{\omega}(i(\omega^2 - \omega_0^2) + \gamma\omega) = r + i\left(m\omega - \frac{k}{\omega}\right)$$

$$\theta = \varphi + \frac{\pi}{2} = -\arg(Z)$$

θ is phase of velocity and φ is phase of position

$$r = \frac{Z_1 - Z_2}{Z_1 + Z_2}$$

$$t = \frac{2Z_1}{Z_1 + Z_2}$$

String and Tension

$$c^2 = \frac{T}{\eta}$$

$$Z = \eta c = \sqrt{T\eta}$$

6 Lens and Diffraction

Snells Law

$$\frac{\lambda_1}{\sin(\theta_1)} = \frac{\lambda_2}{\sin(\theta_2)}$$

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$$

$$n = \frac{c}{v}$$

Lenses

$$\frac{1}{f} = \left(\frac{n_{\text{lens}}}{n_0} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{f} = \frac{1}{o} + \frac{1}{i}$$

$$M = \frac{h'}{h} = -\frac{i}{o}$$

$$\tan(\theta) = \frac{h}{f} \text{ (Parallel Rays)}$$

Michelson Interferometer

$$E_{\text{out}} = rE_2 + tE_1$$

$$t^2 + r^2 = 1$$

$$I_{\text{out}} = |E_{\text{out}}|^2 = 2r^2t^2E_{\text{in}}^2(1 - \cos(\Delta\varphi))$$

$$= 4r^2t^2E_{\text{in}}^2 \sin^2\left(\frac{\Delta\varphi}{2}\right)$$

$$= 4r^2t^2E_{\text{in}}^2 \sin^2\left(\frac{2\pi}{\lambda}(L_2 - L_1)\right)$$

* Note: L_1 and L_2 order does not matter

Abbe's limit/Diffraction

$d = \frac{\lambda}{2n\sin(\theta)}$ or for air

$$d = \frac{\lambda}{2\sin(\theta)}$$

Diffraction

$$d\sin(\theta) = m\lambda$$

$$r = \frac{v_2 - v_1}{v_2 + v_1}$$

$$t = \frac{2v_2}{v_1 + v_2}$$

$$A_t = tA_i$$

$$A_r = rA_i$$

7 Waves and Modes

Standing Waves

Open-Open: $f_n = \frac{nc}{2L}, \quad n = 1, 2, 3, \ldots$

Closed-Open: $f_n = \frac{(2n-1)c}{4L}, \quad n = 1, 2, 3, \ldots$

Closed-Closed: $f_n = \frac{nc}{2L}, \quad n = 1, 2, 3, \ldots$

8 Constants/Miscellaneous

Taylor Approximation

for $|x| \ll 1$

$$\sin(x) \simeq \tan(x) \simeq x$$

$$\cos(x) \simeq 1$$

$$(1+x)^n \simeq (1+xn)$$

Trig Identities

$$A_1\cos(\omega t) + A_2\sin(\omega t) = C\cos(\omega t + \varphi), \quad \tan(\varphi) = -\frac{A_2}{A_1}$$